

Final examination

CSE 420

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Sec: 08

25

## Ans to the Q: 1

Given grammar:

$P \rightarrow MP$

$M \rightarrow \epsilon \quad \{offset = 0\}$

$D \rightarrow Tid; \quad \{Top.put(id.lexeme, T.type, offset);$   
 $offset = offset + T.width;\}$

$D_1$

$D \rightarrow \epsilon$

$T \rightarrow nonid \{ \}$

$D \{ \}$

$T \rightarrow B$

$C$

$B \rightarrow int$

$B \rightarrow float$

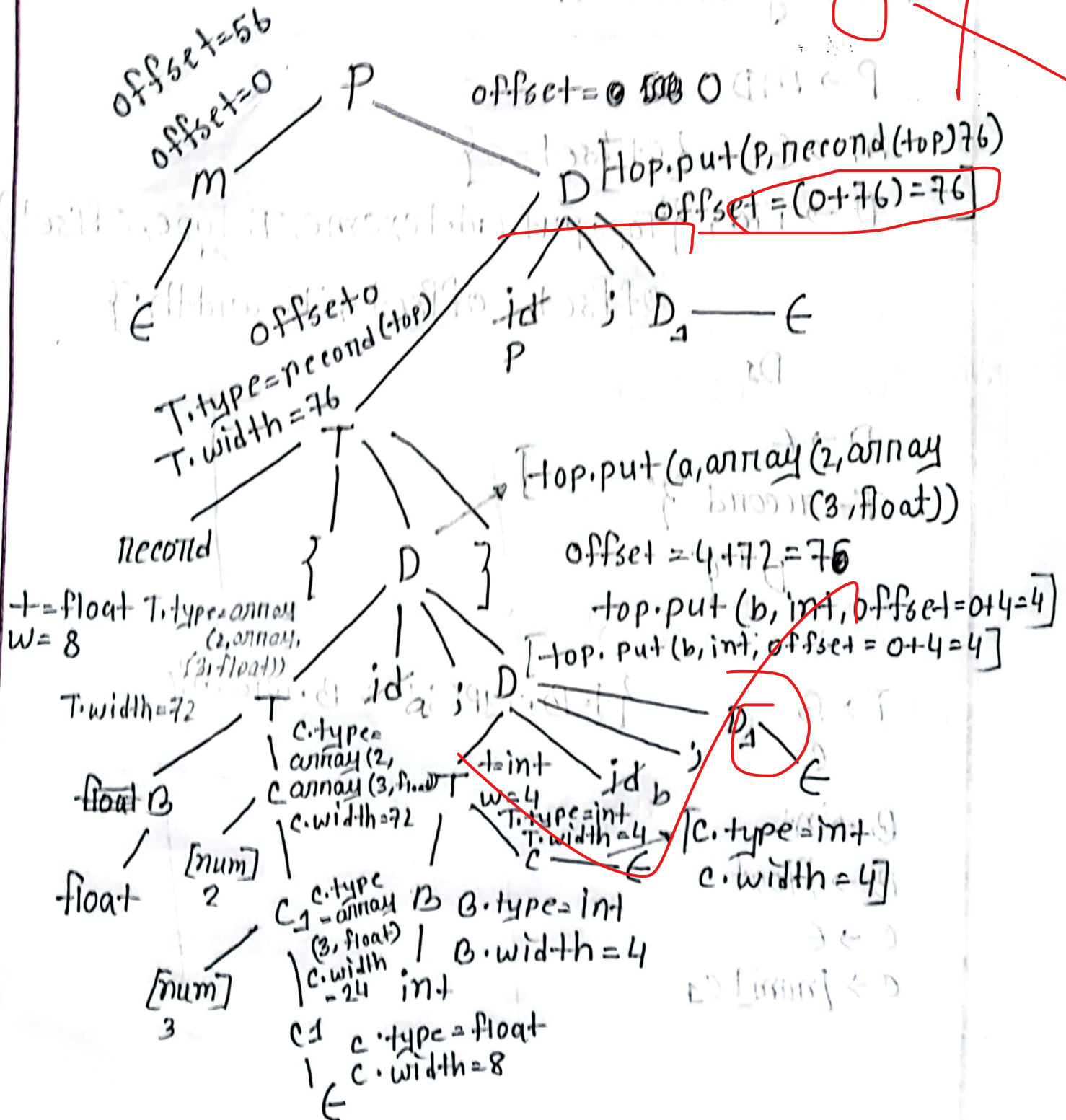
$C \rightarrow \epsilon$

$C \rightarrow [num] C_1$

$\{t = B.type; w = B.width;\}$

1 (a)

```
recond {float [2][3] a; int b;} p;
```



symbol table: & array offset offset

| Variable | Type                      | offset |
|----------|---------------------------|--------|
| a        | array(2, array(3, float)) | 0      |
| b        | int                       | 48     |
| p        | int                       | 52     |

offset = 56

$$0 + 1T = 1T \quad (1)$$

$$1T + 1T = 2T$$

$$2T + 1T = 3T$$

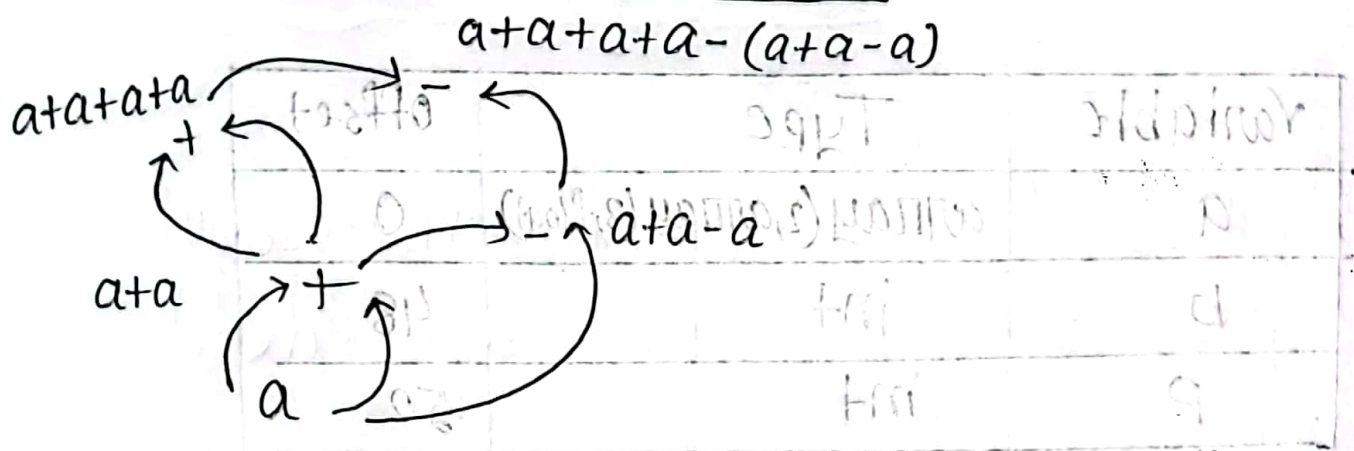
$$3T + 1T = 4T$$

② Examples and triple representation

the three values are

below:

Ans to the q: no 3: a) dot notation



①  $T_1 = a+a \quad 0$

$T_2 = T_1 - a \quad 1$

$T_3 = T_1 + T_1 \quad 2$

$T_4 = T_3 - T_2 \quad 3$

② Quadriples and triples representation of the three address code are given below:

\* Quad tuples:

|   | op | arg1           | arg2           | result         |
|---|----|----------------|----------------|----------------|
| 0 | +  | a              | a              | T <sub>1</sub> |
| 1 | -  | T <sub>1</sub> | a              | T <sub>2</sub> |
| 2 | +  | T <sub>1</sub> | T <sub>1</sub> | T <sub>3</sub> |
| 3 | -  | T <sub>3</sub> | T <sub>2</sub> | T <sub>4</sub> |

\* Triples:

|   | op | arg1 | arg2 |  |
|---|----|------|------|--|
| 0 | +  | a    | a    |  |
| 1 | -  | (0)  | a    |  |
| 2 | +  | (0)  | (0)  |  |
| 3 | -  | (2)  | (1)  |  |

Ans to the q: no2

Here, active symbol table by using  
hash-table where  $n = r * m$

